[INCH-POUND]
A-A-59244
June 16, 1998
SUPERSEDING
OO-D-2776
November 12, 1992

COMMERCIAL ITEM DESCRIPTION

DRILL, PNEUMATIC DRIFTER, SELF-PROPELLED, CRAWLER MOUNTED (TRACK DRILL)

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers two types of self-propelled, crawler mounted, drifter drills (referred to hereinafter as "track drill"), having not less than a 4.75-inch (121 millimetre (mm)) cylinder bore. The track drills are used for horizontal and vertical drilling to define crevice lines, to drill blast holes, and to accomplish other miscellaneous drilling jobs.

2. CLASSIFICATION. The track drills covered by this CID are classified by the method utilized for rotation of the pneumatic drifter drill and are classified as specified (see 7.2):

Type I - Rifle bar.
Type II - Independent air motor drive.

3. SALIENT CHARACTERISTICS.

3.1 Description. The track drill shall consist of a crawler chassis with a pintle attachment for towing an air compressor, along with the following:

a. Air motors.
b. Dust suppression system.
c. Hydraulically-operated swing type boom.
d. Chain-fed type drill retainer channel.
e. Either an independent air motor or rifle bar rotation type pneumatic drifter drill.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3820

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The track drill shall be provided with lubrication fittings, hoses, controls, crawler brakes or locks, and other necessary items to perform complete and efficient drilling operations. The track drill shall use compressed air from the towed air compressor as a power source for propulsion and drilling.

3.2 Standard commercial product. The track drills shall, as a minimum, be in accordance with the requirements of this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product, shall be included in the track drill being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Interchangeability. All track drills of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Design and construction.

3.4.1 Safety. Except for the drill steel, feed chain, and crawler track assembly, all rotating or reciprocating parts that are hazardous to operating personnel shall be enclosed or guarded.

3.4.2 Noise warning plate. A warning plate made of corrosion-resistant material, which will last and remain legible for the life of the equipment, shall be securely affixed with nonferrous screws, rivets, or bolts to a location conspicuous from the control panel. The plate shall have a yellow background with black lettering inscribed as follows:

CAUTION
EAR PROTECTION REQUIRED
WHEN EQUIPMENT IS OPERATING

3.4.3 Ease of maintenance. All major assemblies and installed attachments shall be readily accessible for servicing, repair, or replacement. Drainage of lubricants and hydraulic fluid shall be to the ground without draining on or over any part of the track drill. When drain outlets are located over parts of the track drill, integral tubes or pipes shall convey the fluid from the drain to the ground. Drain outlets shall be accessible without the removal or adjustment of accessories or parts other than access opening covers.

3.4.4 Crawler chassis. The track drill shall be mounted on a self-propelled crawler chassis consisting of two oscillating crawler tracks. The crawler chassis shall have a ground clearance of not less than 10 inches (254 mm). The crawler mounting shall permit tracks to rise over obstructions while maintaining a proportionately level platform when traveling or drilling. The vertical oscillation of the track frame shall be not less than 14 inches (356 mm), measured at the center of the front idler. Heavy-duty recoil springs shall be furnished for cushioning front idlers. Track rollers and the track rail shall be arranged to prevent damage to bearings or grease fittings

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when the track drill is operated through abrasive material or loose stone. The track drill shall be equipped with track adjusters or maintaining tension on the track. Track frames shall not be spliced. The track gage shall be not less than 70 inches (1 778 mm) from center to center of tracks and a ground bearing length of not less than 60 inches (1 524 mm).

3.4.4.1 Track chain. Train links and grouser shoes shall be heat-treated steel of drop-forged or roller construction. The shoes shall be not less than 10 inches (254 mm) wide with provisions for affixing rubber crawler-tread pads. Track links shall be equipped with replaceable pins and bushings. The master pin shall be secured by a lock or a press fit and shall be identified.

3.4.4.2 Tramming air motors. A tramming air motor using a gear drive train or chain drive, shall drive each track. Each motor shall have a spring-loaded deadman control for forward and reverse movement of the track. The motor and drive train shall be enclosed. Adequate vent plugs shall be provided to prevent air pressure build-up in track drive housing. The rated horsepower (watt (W)) of the air motors at 90 pound-force per square inch (psi) (621 kilopascal (kPa)) pressure shall be not less than that required to advance the track drill up a grade of not less than 20 percent for 100 yards (91 metre (m)), and over level ground at a speed of not less than 1.3 miles per hour (2 kilometre per hour (km/h)). These requirements are when towing the air compressor unit weighing not less than 8,500 pounds (lb) (3 856 kilogram (kg)). The air motors shall have means for being disengaged from the crawler driven train without the use of tools to permit the track drill to be easily towed.

3.4.4.3 Brakes. Each track shall be provided with brakes or track locks to prevent movement of the track drill when on a 50 percent longitudinal slope. Track locks or brakes shall be operable when power is lost.

3.4.4.4 Platform. When specified (see 7.2), a platform shall be furnished for use by the track drill operator while manipulating the tramming controls during road travel.

3.4.4.5 Toolbox. A lockable metallic toolbox shall be provided to carry those tools and accessories required to operate the unit. The toolbox shall be securely fastened to the chassis in a protected and accessible location.

3.4.4.6 Pintle. A pintle assembly shall be attached to the chassis to receive the tongue of a 600 or 750 cubic feet per minute (cfm) (0.3 or 0.4 cubic metre per second (m^3/s)) air compressor.

3.4.5 Boom. The boom shall be hydraulically raised, lowered, and swung. Vertical movements of the boom shall be such that holes may be drilled anywhere within the range of from 2.50 to 8 feet (0.8 to 2.4 m) above the ground surface of the tracks. Movement of the boom shall be such that holes may be drilled on a line not less than 12 inches (305 mm) outside of either track without repositioning the crawler. The pneumatic drifter drill of the track drill shall be capable of drilling regardless of the position of the boom and mast. Boom movement shall be so controlled that the boom can be stopped and fixed without creepage at all points in its range of vertical travel.

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3.4.6 Mast. A mounting plate shall be used to secure the fabricated steel mast to the boom. The mounting plate shall be hydraulically-operated and shall be capable of sliding on the mast for a distance of not less than 3 feet (0.9 m). A reversible air motor or hydraulic cylinder(s), secured to the mast, shall be used to drive the feed mechanism. The mast assembly shall include an adjustable feed chain, upper and lower feed chain idler sprockets, a pneumatic drifter drill mounting plate, a foot pad, and a drill steel centralizer. Feed chain and idler sprockets shall be arranged so that total travel of the pneumatic drifter drill shall allow the operator to change 12-foot (3.7 m) sectional drill steel and couplings, by disengagement of the centralizer only. When operating at an air pressure of 90 psi (621 kPa), the air motor or hydraulic cylinder(s) shall drive the feed chain, and shall provide an effort of not less than 3,000 lb (1 361 kg) for crowding and retracting the pneumatic drifter drill. The mast assembly shall be capable of a hydraulic roll operation through an arc of not less than 70 degrees from right to left. The vertical mast dump from a horizontal position shall have a travel of not less than 145 degrees with not more than one mechanical pin change. The mast movement shall be controlled so that the mast can be stopped and fixed without creepage at all points in its range of travel.

3.4.7 Pneumatic drifter drill. The pneumatic drifter drill shall operate on a rated air pressure of 90 psi (621 kPa), and shall be automatically valved to provide not less than 1,500 blows per minute to the chucked drill shank. The pneumatic drifter drill for the type I track drill shall weigh not less than 160 lb (73 kg) and shall be of the long stroke hammer type. The pneumatic drifter drill for the type II track drill shall weigh not less than 280 lb (127 kg) and shall be of the high speed, short stroke type with independent rotation. The chuck shall be designed to prevent the shank from leaving the pneumatic drifter drill, and shall contain parts of high wearing quality. The chuck shall accommodate a four spline drill shank with rope thread. Unless otherwise specified (see 7.2), the pneumatic drifter drill shall be furnished with five drill shanks. The pneumatic drifter drill shall also provide for control of clockwise, neutral, and counterclockwise rotation of the drill rod. The pneumatic drifter drill for the type II track drill shall also provide drill rod rotation independent of hammer control. The pneumatic drifter drill shall be designed for wet or dry drilling, and to allow water or air to pass through the unit and the attached drill rod for flushing or blowing purposes. A lubricator shall be provided in the air line and have an oil reservoir of sufficient capacity for not less than 8 hours of operation. The reservoir shall be equipped with a filler opening with a removable strainer. The cap or reservoir immediately adjacent to the filler cap shall be plainly marked to indicate “Rock Drill Oil.” The lubricator shall give thorough and positive lubrication to the pneumatic drifter drill and feed motor at all times regardless of the position of the pneumatic drifter drill.

3.4.8 Hydraulic system. The hydraulic system shall be protected against the entrance of dirt or other foreign materials. The hydraulic system shall be complete with all operating accessories, including a pump, which is driven by an air motor. Hydraulic cylinders shall be of the double-acting type having lock-type check valves. Hydraulic valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. A wiper shall be provided at each end of the cylinders. A wiper shall be provided on the cylinders, adjacent to the piston rod seal, to prevent dirt and abrasives from damaging the seal or entering the cylinder.

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3.4.9 Controls. All controls shall be clearly marked as to use and function. The crawler, boom, and mast controls shall be grouped for safe operation and ease of operation when positioning the crawler, boom, and mast preparatory to drilling. The drilling controls shall be grouped for safe operation and shall be accessible to the operator when drilling at any position. Drilling controls shall operate the functions of the pneumatic drifter drill, including the rotation, feed, and flushing operations.

3.4.10 Hoses. The quantity of interconnecting hose lines shall be held to a minimum, consistent with complete operation and shall be provided with nonleak connections. Hoses shall be free from pits, blisters, checks, or other defects and shall be of uniform thickness. Hoses shall be designed for not less than 200 psi (1 379 kPa) working pressure and shall pass a 600 psi (4 137 kPa) hydrostatic test pressure. Pneumatic hose shall have an oil-resistant inner tube and scuff-resistant outer cover. Pneumatic hose shall be supported, where practicable, to prevent kinking. One connection shall be provided to receive the supply hose line from the compressor and shall be 2 inches (51 mm) with 11.50 threads per inch National Taper Pipe (thread), external thread.

3.4.11 Dust-suppression system. Unless otherwise specified (see 7.2), a dust-suppression system shall be provided. The dust-suppression system shall be protected against the entrance of dirt or other foreign materials. It shall be complete with all operating accessories, including a suitable steel reservoir, automatic air operated valve that turns on liquid flow when blowing air is turned on, adjustable fine control, a filler-opening cap, and hoses. The steel reservoir shall be of sufficient size to hold the quantity of detergent water solution necessary to ensure 8 hours of operation of the equipment without being refilled. There shall be no leakage of fluid when the system is operated.

3.4.12 Air transportability configuration. When specified (see 7.2), the track drill shall be transportable by aircraft in a roll-on/roll-off configuration in accordance with MIL-STD-1791. The air transportability configuration shall be for air transport by C-5, C-17, C-130, or C-141 aircraft as specified (see 7.2).

3.4.13 Rail impact. When specified (see 7.2), the track drill shall meet the rail impact requirements of MIL-STD-810.

3.5 Lifting and tiedown attachments. When specified (see 7.2), the track drill shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the track drill. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the track drill on the carrier when shipped.

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3.6 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 7.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current process and the total dry film thickness shall be not less than 2.5 mil (0.0635 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.7 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J 534. Where use of high-pressure lubricating equipment, 1,000 psi (6 895 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.8 Identification plate. An identification plate will be furnished by the contracting officer for each track drill. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each track drill in a conspicuous place with nonferrous screws, rivets, or bolts. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.9 Instruction plates. The track drill shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws rivets, or bolts. A warning plate shall be conspicuously affixed to the unit immediately adjacent to the reservoir filler cap of the air line lubricator. The plate shall be inscribed as follows:

WARNING

Under normal operation, the line oiler system is pressurized to 90 psi (621 kPa). Do not open filler cap until air pressure has been completely eliminated from the air system.

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3.10 Performance.

3.10.1 Drilling. The following drilling requirements shall be met with the pneumatic drifter drill employing a button-type bit. The rated operating air pressure shall be 90 psi (621 kPa). The rate of use of the air shall be not greater than 600 cfm (0.3 m^3/s) for type I track drill and 750 cfm (0.4 m^3/s) for type II.

3.10.1.1 Drilling in Barre granite. The pneumatic drifter drill shall drill a 3.50-inch (89 mm) diameter hole, 12 inches (305 mm) deep in not less than 1 minute when operating in Barre granite or equal. The granite shall have a crushing strength of not less than 17,000 psi (117 211 kPa).

3.10.1.2 Drilling in homogeneous rock. The pneumatic drifter drill shall drill 30-foot (9 m) deep holes in homogeneous rock with an average specific gravity of not less than 2.6. The boom and mast shall be at various angles and positions during drilling. Drilling shall be performed for 50 hours of throttle time without replacement of parts other than drill steel and bits.

3.10.2 Stability. The track drill shall operate on 35 percent side slopes, or 50 percent longitudinal slopes, or any combination thereof without either track losing all contract with the ground.

3.11 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of IEEE SI-10 and all other requirements of this commercial item description including form, fit and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

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7. NOTES.

7.1 Source of documents.

7.1.1 Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

7.1.2 The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

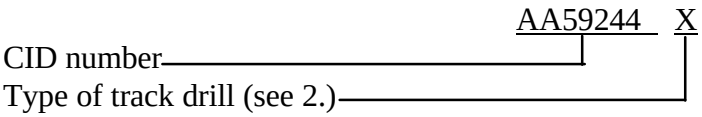
7.1.3 IEEE Standard is available form the Institute of Electrical and Electronics Engineers, IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.1.4 SAE Standards and Recommended Practices are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Type of track drill required (see 2.).
c. When an operator's platform is required (see 3.4.4.4).
d. Number of drill shanks required, if other than five (see 3.4.7).
e. When a dust-suppression system is not required (see 3.4.11).
f. When track drill is to be configured for air transport. Type of aircraft for which track drill is to be configured (see 3.4.12).
g. When the track drill is to meet the rail impact test of MIL-STD-810 (see 3.4.13).
h. When lifting and tiedown attachments are required (see 3.5).
i. Color of finish coat (see 3.6).

7.3 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN used for items acquired to this description are created as follows:



PIN codes:

1 = Type I
2 = Type II

7.4 Supersession data. This commercial item description replaces Federal Specification OO-D-2776, dated November 12, 1992.

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7.5 Classification cross reference. Classifications used in this commercial item description (see 2.) are identical to those found in the superseded Federal Specification OO-D-2776.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Army - MT
Navy - MC
Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 3820-0031)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59244	**2. DOCUMENT DATE (YYMMDD)** 980616

3. DOCUMENT TITLE

DRILL, PNEUMATIC DRIFTER, SELF-PROPELLED, CRAWLER MOUNTED (TRACK DRILL)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER (CODE 15E2), 4111 SAN PEDRO ST. PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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